

Mosses and *Nostoc*: an update from the Fichtel Mountains

Martin Feulner^{1,2}, Ulla Spieleder¹, Frederic J. Pascual
Dose³, Jona Motzkus¹, Daniel C. Thomas^{2,3}

¹University of Bayreuth Plant Systematics Working Group ²Chair of Microbial Ecology, University of Bayreuth ³BayCEER - Bayreuth Center of Ecology and Environmental Research

Introduction

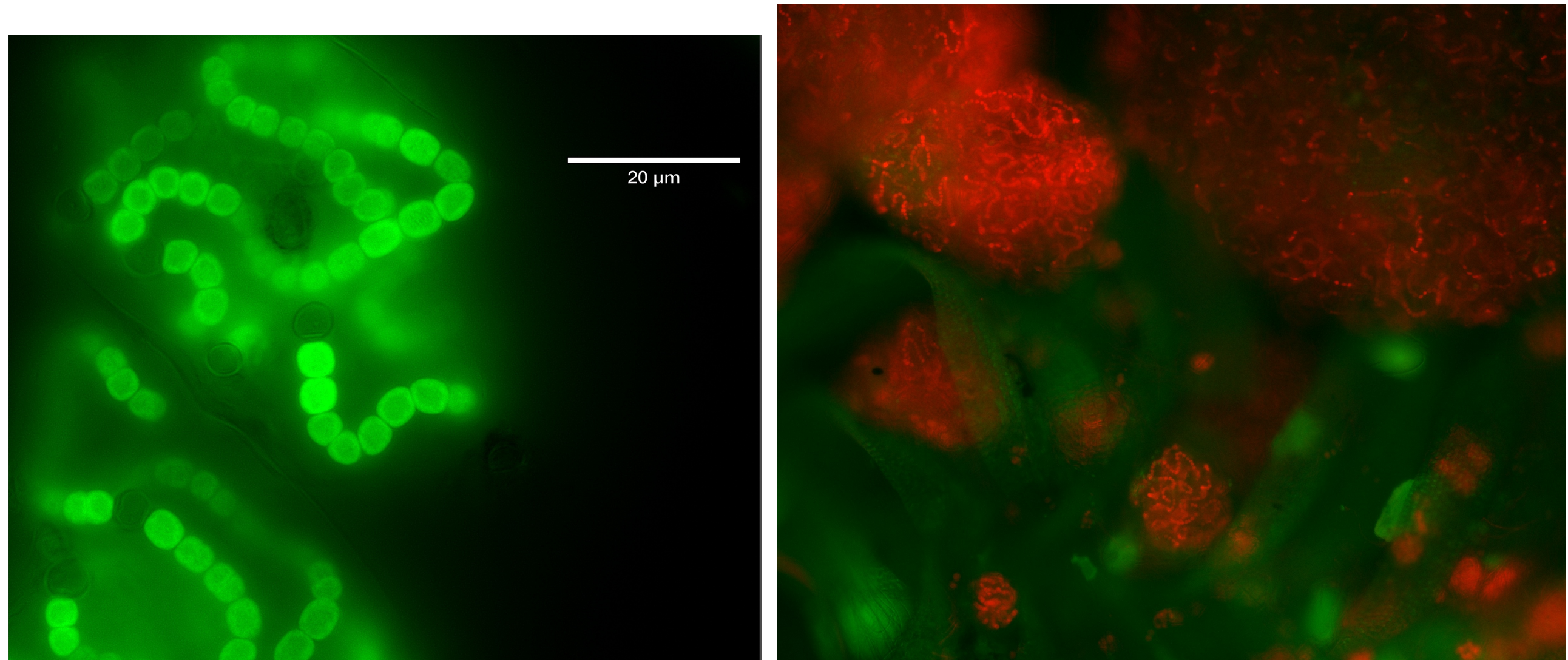
- Mosses are hotspots of nitrogen-fixation, due to diazotrophic symbionts in their microbiomes.
- Because of the large scale ecological service provided by these microbes, moss-cyanobacterial symbioses are of increasing interest.
- Historically the most commonly studied such symbioses are *Nostoc* spp. with "feather" (Hypnales) and *Sphagnum* mosses.
- Most studies have occurred in Scandinavia and the North American Boreal forest zone.



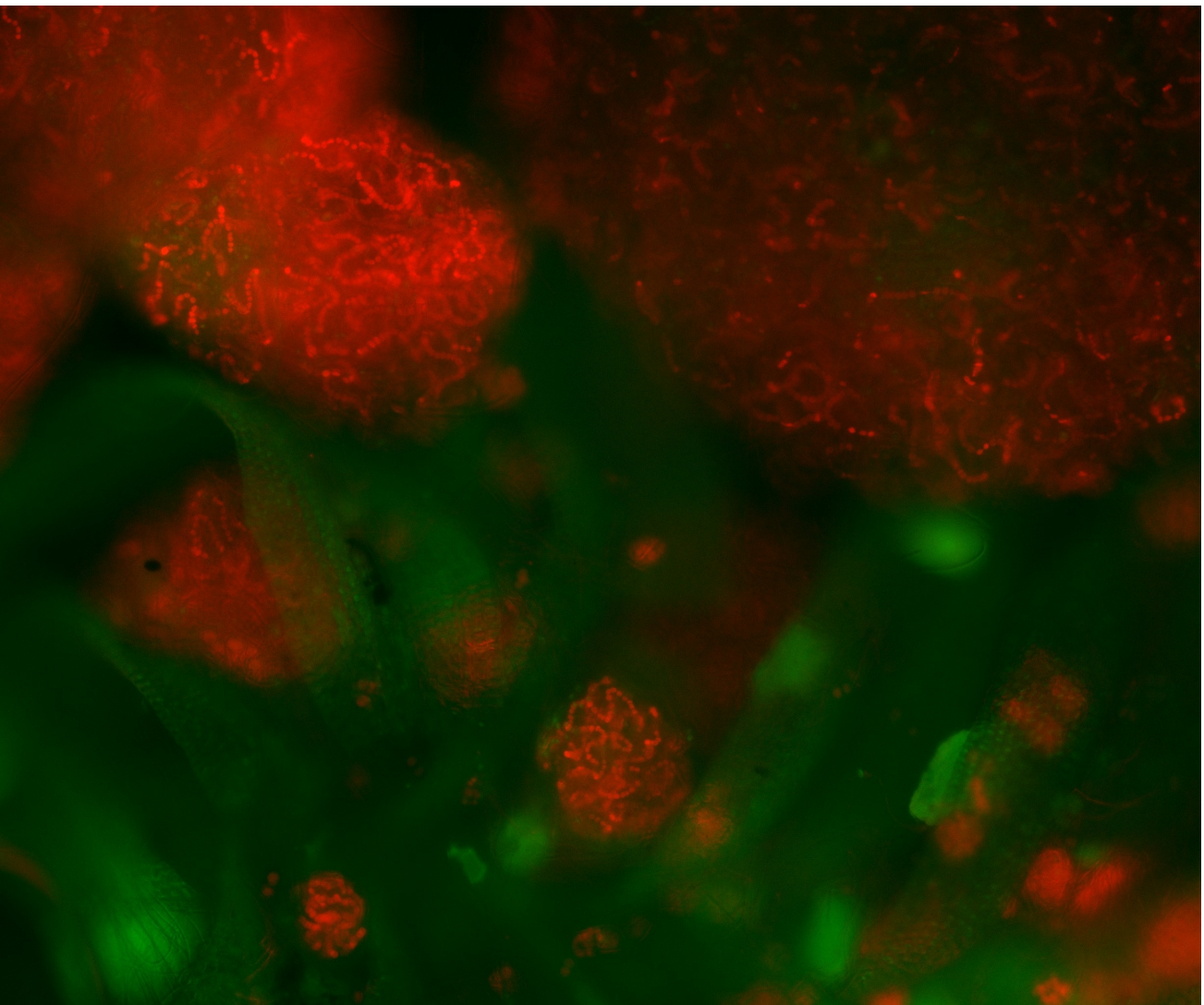
Waldstein forest, a typical field site in the Fichtelgebirge mountains.



Hylocomium splendens moss with *Nostoc* colonies. Drawing by Jona Motzkus (2024).



Nostoc sp. cultured from *Hylocomium* moss, forest floor.



Compactionostoc sp. cultured from *Zygodon dentatus* moss.

Hypotheses

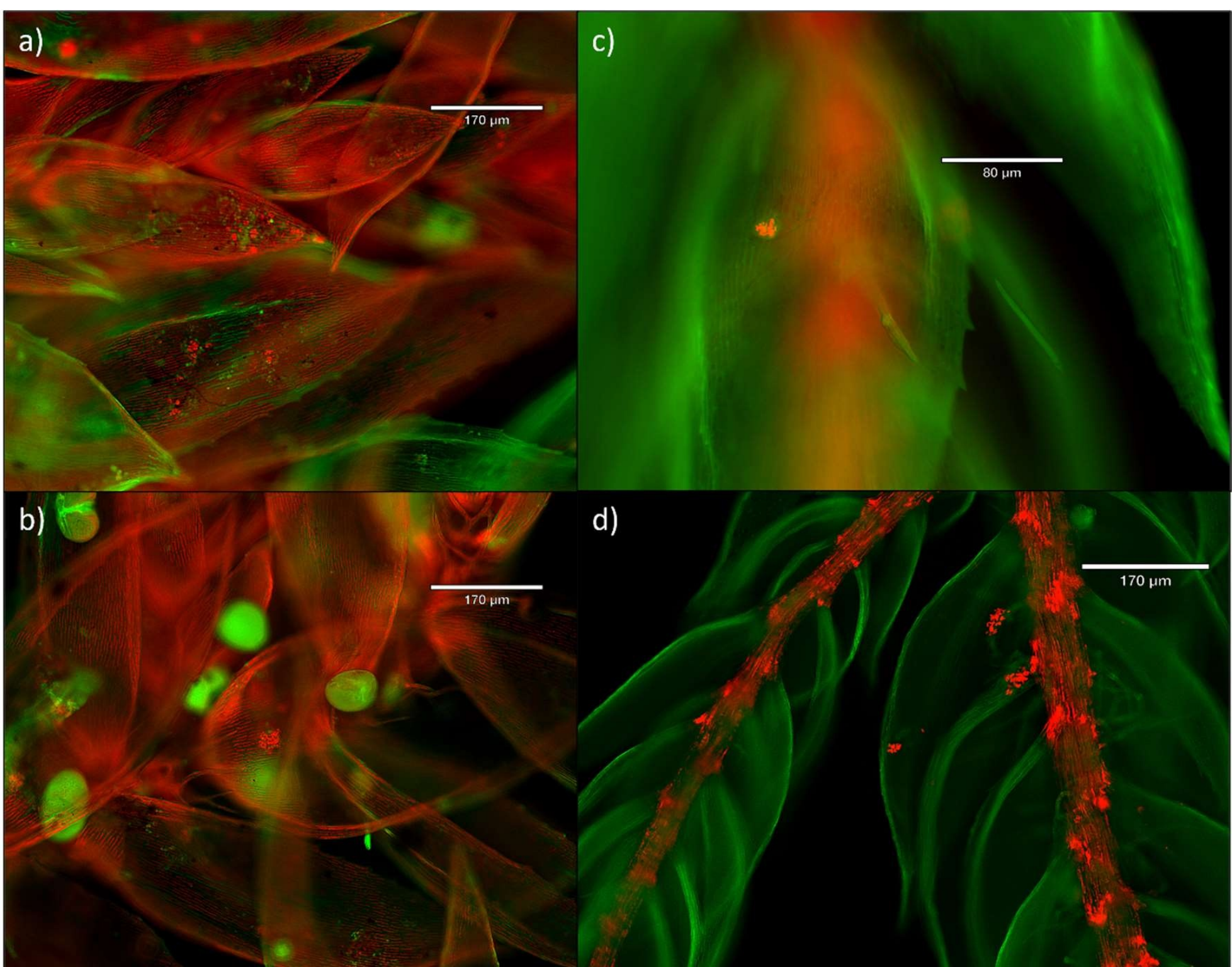
- Moss-*Nostoc* symbioses are "loose", better predicted by environmental conditions than host moss phylogeny.
- Mosses in the forest canopy are under selection pressure to enter into symbioses with nitrogen-fixing bacteria due to ephemeral environmental nitrogen sources.

Methods

- Field surveys for *Nostoc*-like colonies in mosses, especially rare arboreal mosses
- Light and epi-fluorescence microscopy
- attempted culturing of all *Nostoc*-like organisms
- chemical surface sterilization for detection of endophytes
- colony PCR of *NifH* and full-length 16S
- nanopore sequencing



Results



- Possible endophytic colonization by cyanobacteria of stems of mosses was observed.

- Arboreal mosses *Orthotrichum* and *Zygodon* (Orthotrichaceae) were frequent hosts to *Nostoc*-like organisms, possibly indicating a specialized niche for the symbiosis.

- A new moss host family was observed (Orthotrichaceae) extending the range of known moss hosts for *Nostoc*-like organisms beyond feather mosses and *Sphagnum*.

- Multiple alternate *nifH* genes were observed within the genomes of some *Nostoc*-like organisms, possibly enabling diazotrophy under variable ecological conditions.

- moss-associated *Nostoc*-like organisms were positioned throughout our 16S Nostocaceae phylogeny, with no obvious single moss-symbiotic clade in the family.

